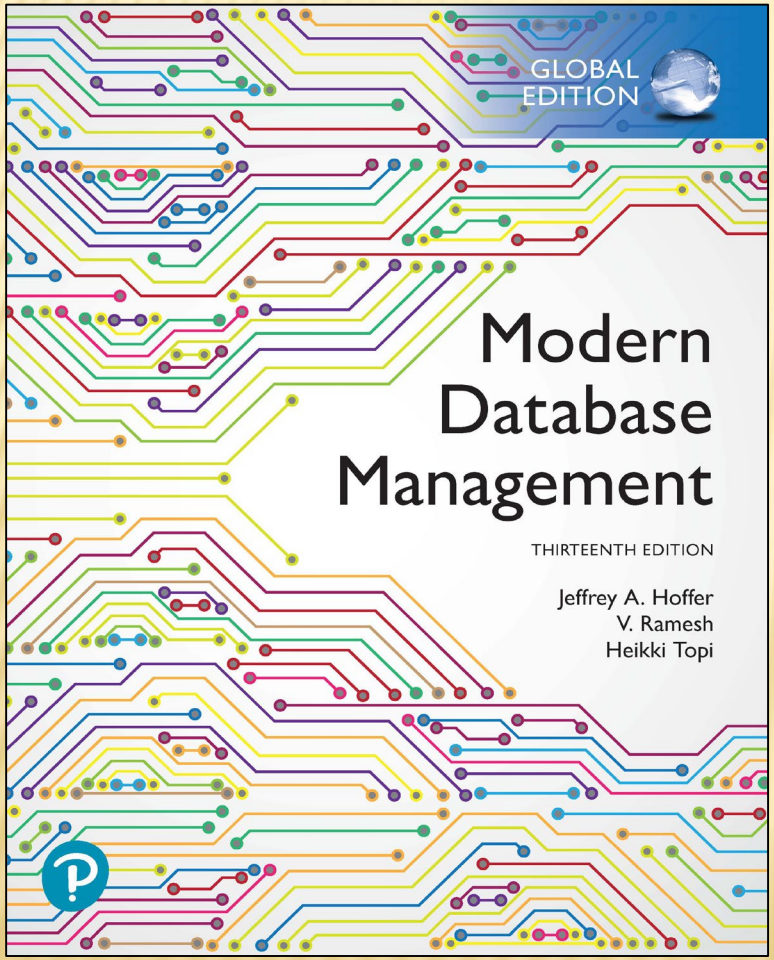
Modern Database Management

Thirteenth Edition



Chapter 1

The Database Environment and Development Process

Learning Objectives (1 of 2)

1.1 Define terms

1.2 Name limitations of conventional file processing

1.3 Explain advantages of databases

1.4 Identify costs and risks of databases

1.5 Distinguish between operational and informational data management

1.6 List components of database environment

Learning Objectives (2 of 2)

1.7 Identify categories of database applications

1.8 Explain prototyping and agile development approaches

1.9 Explain roles of individuals

1.10 Explain differences between personal, multi-tiered, and enterprise data management

1.11 Explain three-schema architectures (external, conceptual, internal)

Definitions

- **Database:** organized collection of logically related data
- **Data:** stored representations of meaningful objects and events
  - Structured: numbers, text, dates 結構化
  - Unstructured: images, video, documents 非結構化
- **Information:** data processed to increase knowledge in the person using the data
- **Metadata:** data that describes the properties and context of user data

Figure 1-1 Converting Data To Information (1 OF 2)

(a) Data in context

Class Roster

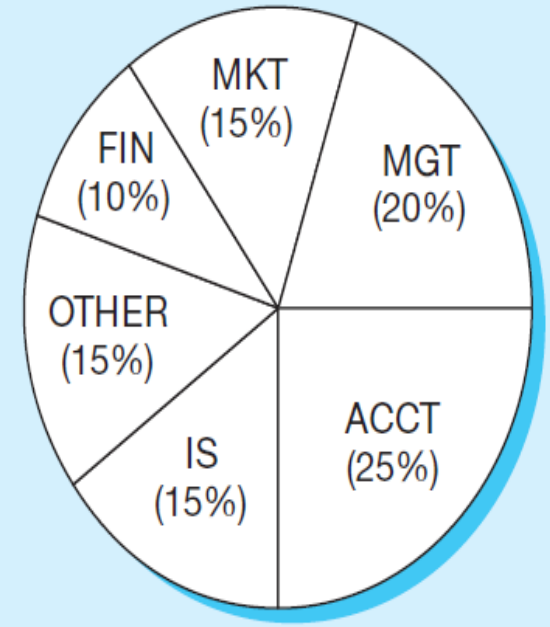
Course: MGT 500 Business Policy Semester: Spring 2018

Section: 2

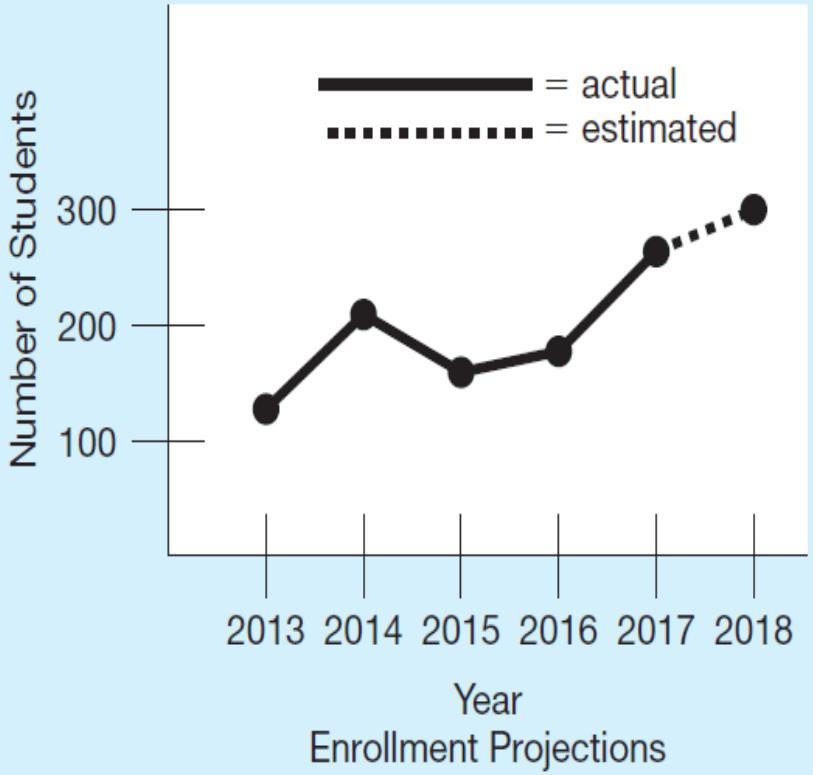
Name	ID	Major	GPA
Baker, Kenneth D.	324917628	MGT	2.9
Doyle, Joan E.	476193248	MKT	3.4
Finkle, Clive R.	548429344	PRM	2.8
Lewis, John C.	551742186	MGT	3.7
McFerran, Debra R.	409723145	IS	2.9
Sisneros, Michael	392416582	ACCT	3.3

Figure 1-1 Converting Data To Information (2 OF 2)

(b) Summarized data



Percent Enrollment by Major (2018)



Enrollment Projections

Table 1-1 Example Metadata For Class Roster

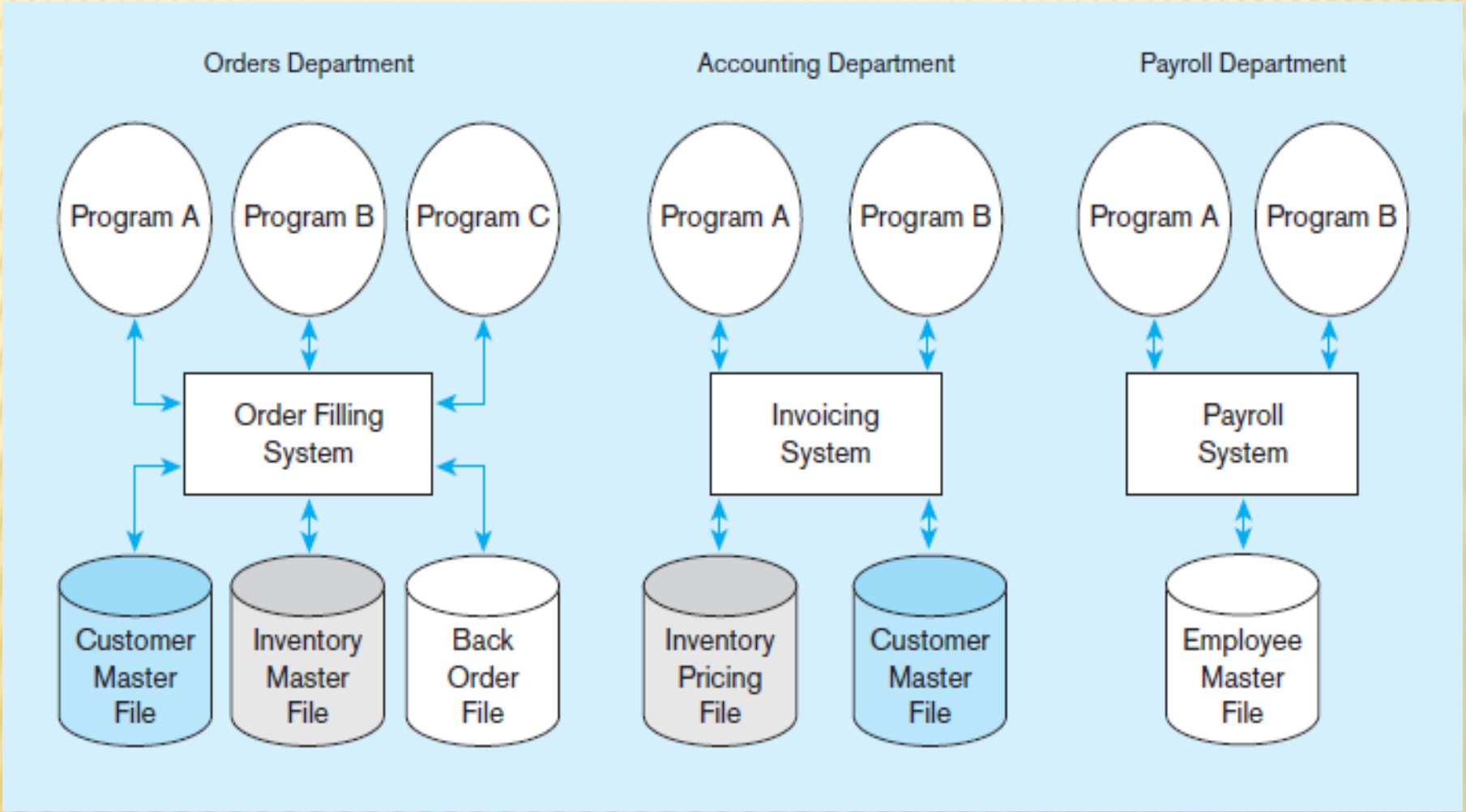
- Descriptions of the properties or characteristics of the data, including data types, field sizes, allowable values, and data context

Name	Type	Length	Min	Max	Description	Source
Course	Alphanumeric	30	Blank	Blank	Course ID and name	Academic Unit
Section	Integer	1	1	9	Section number	Registrar
Semester	Alphanumeric	10	Blank	Blank	Semester and year	Registrar
Name	Alphanumeric	30	Blank	Blank	Student name	Student IS
ID	Integer	9	Blank	Blank	Student ID (SSN)	Student IS
Major	Alphanumeric	4	Blank	Blank	Student major	Student IS
GPA	Decimal	3	0.0	4.0	Student grade point average	Academic Unit

Disadvantages Of File Processing

- Program-Data Dependence
  - All programs maintain metadata for each file they use
- Duplication of Data
  - Different systems/programs have separate copies of the same data
- Limited Data Sharing
  - No centralized control of data
- Lengthy Development Times
  - Programmers must design their own file formats
- Excessive Program Maintenance
  - 80% of information systems budget

Figure 1-2 Old File Processing Systems at Pine Valley Furniture Company



The Database Approach (1 of 2)

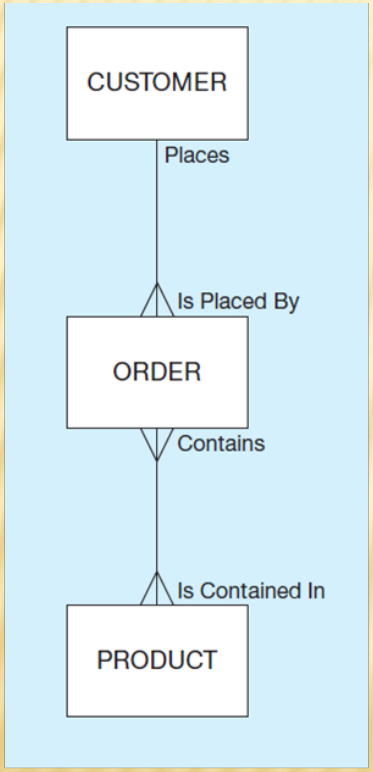
- Data models
  - Graphical diagram capturing nature and relationship of data
  - Enterprise Data Model – high-level entities and relationships for the organization
  - Project Data Model – more detailed view, matching data structure in database or data warehouse
- Entities 實體
  - Noun form describing a person, place, object, event, or concept
  - Composed of attributes 屬性
- Relationships 關聯
  - Between entities

ER Model = Entity + Relationships

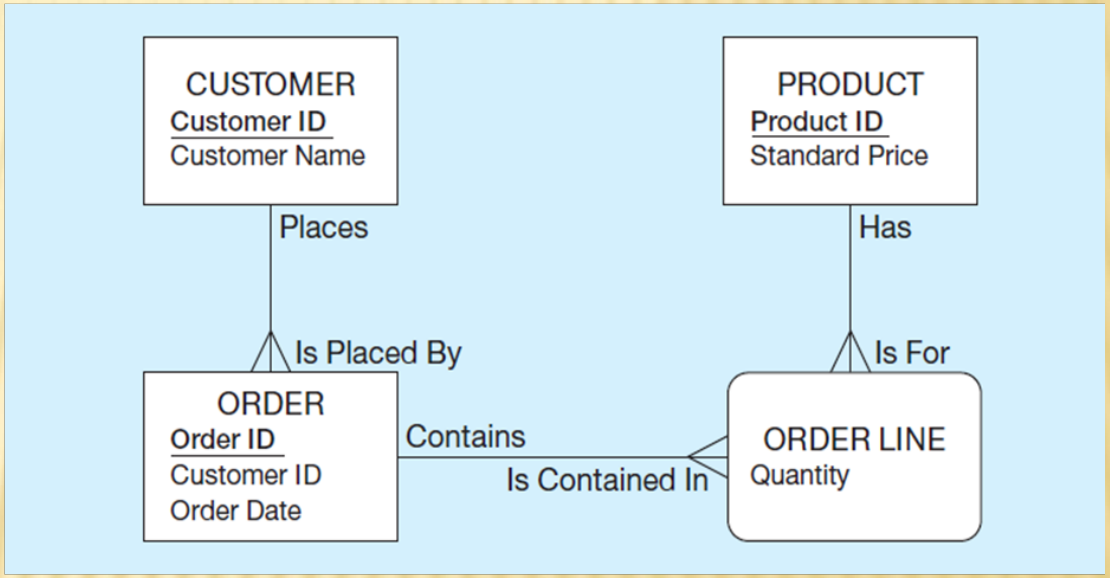
+ Usually one-to-many (1:N) or many-to-many (M:N), but can also be one-to-one (1:1)

Figure 1-3 Comparison of Enterprise- and Project-Level Data Models

(a) Segment of an enterprise data model



(b) Segment of a project data model

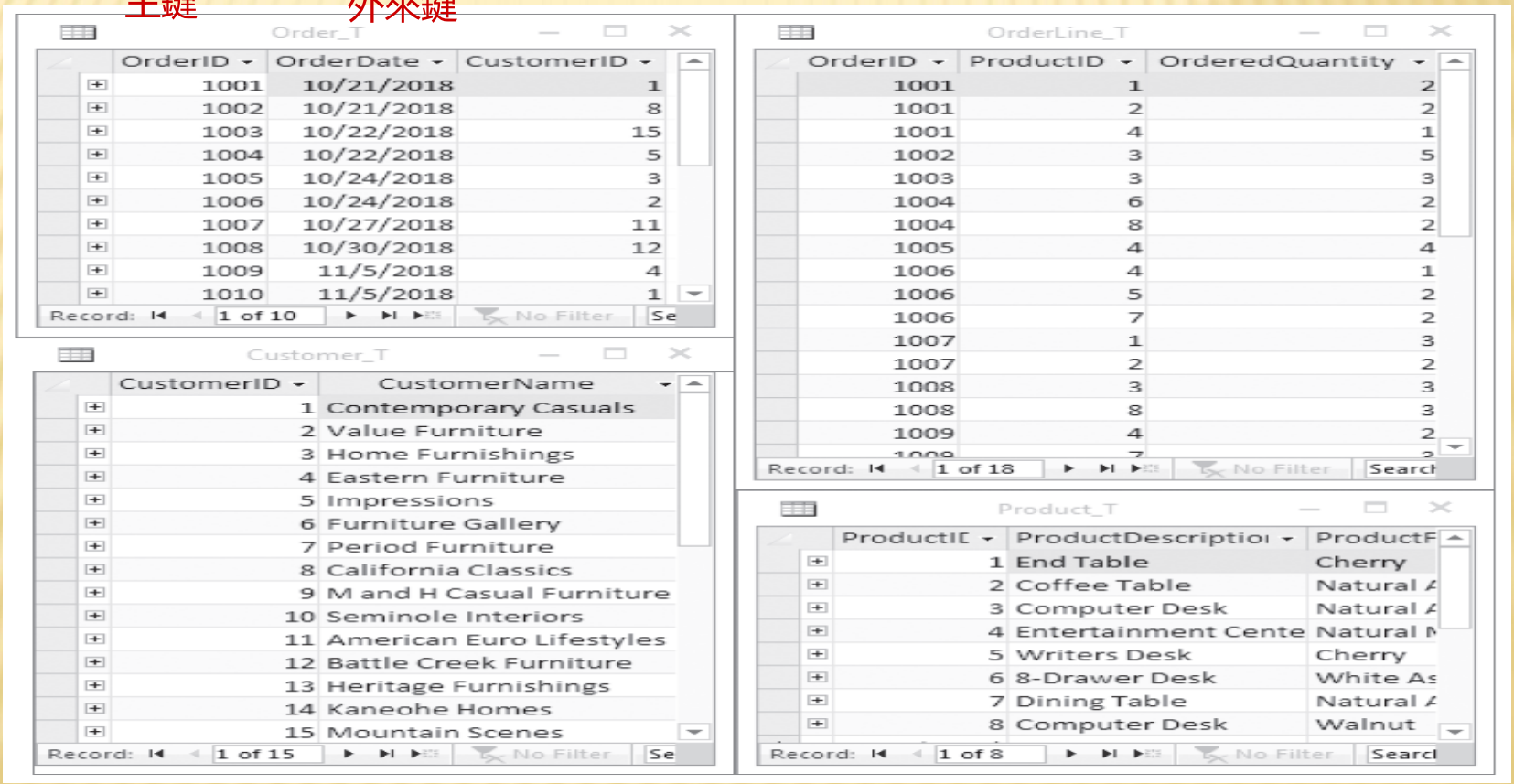


The Database Approach (2 OF 2)

關聯表

- Relational Databases
- Database technology involving tables (relations) representing entities and primary/foreign keys representing relationships (see Figure 1-17)

主鍵　　外來鍵



Order_T

OrderID	OrderDate	CustomerID
1001	10/21/2018	1
1002	10/21/2018	8
1003	10/22/2018	15
1004	10/22/2018	5
1005	10/24/2018	3
1006	10/24/2018	2
1007	10/27/2018	11
1008	10/30/2018	12
1009	11/5/2018	4
1010	11/5/2018	1

Record: 1 of 10

OrderLine_T

OrderID	ProductID	OrderedQuantity
1001	1	2
1001	2	2
1001	4	1
1002	3	5
1003	3	3
1004	6	2
1004	8	2
1005	4	4
1006	4	1
1006	5	2
1006	7	2
1007	1	3
1007	2	2
1008	3	3
1008	8	3
1009	4	2

Record: 1 of 18

Customer_T

CustomerID	CustomerName
1	Contemporary Casuals
2	Value Furniture
3	Home Furnishings
4	Eastern Furniture
5	Impressions
6	Furniture Gallery
7	Period Furniture
8	California Classics
9	M and H Casual Furniture
10	Seminole Interiors
11	American Euro Lifestyles
12	Battle Creek Furniture
13	Heritage Furnishings
14	Kaneohe Homes
15	Mountain Scenes

Record: 1 of 15

Product_T

ProductID	ProductDescription	ProductF
1	End Table	Cherry
2	Coffee Table	Natural A
3	Computer Desk	Natural A
4	Entertainment Cente	Natural M
5	Writers Desk	Cherry
6	8-Drawer Desk	White As
7	Dining Table	Natural A
8	Computer Desk	Walnut

Record: 1 of 8

簡答題第一題：請解釋什麼是 Primary keys

Advantages Of The Database Approach

- Program-data independence
- Planned data redundancy
- Improved data consistency
- Improved data sharing
- Increased application development productivity
- Enforcement of standards
- Improved data quality
- Improved data accessibility and responsiveness
- Reduced program maintenance
- Improved decision support

Costs And Risks Of The Database Approach

- New, specialized personnel
- Installation and management cost and complexity
- Conversion costs
- Need for explicit backup and recovery
- Organizational conflict

Figure 1-5 Integrated Data Management Framework

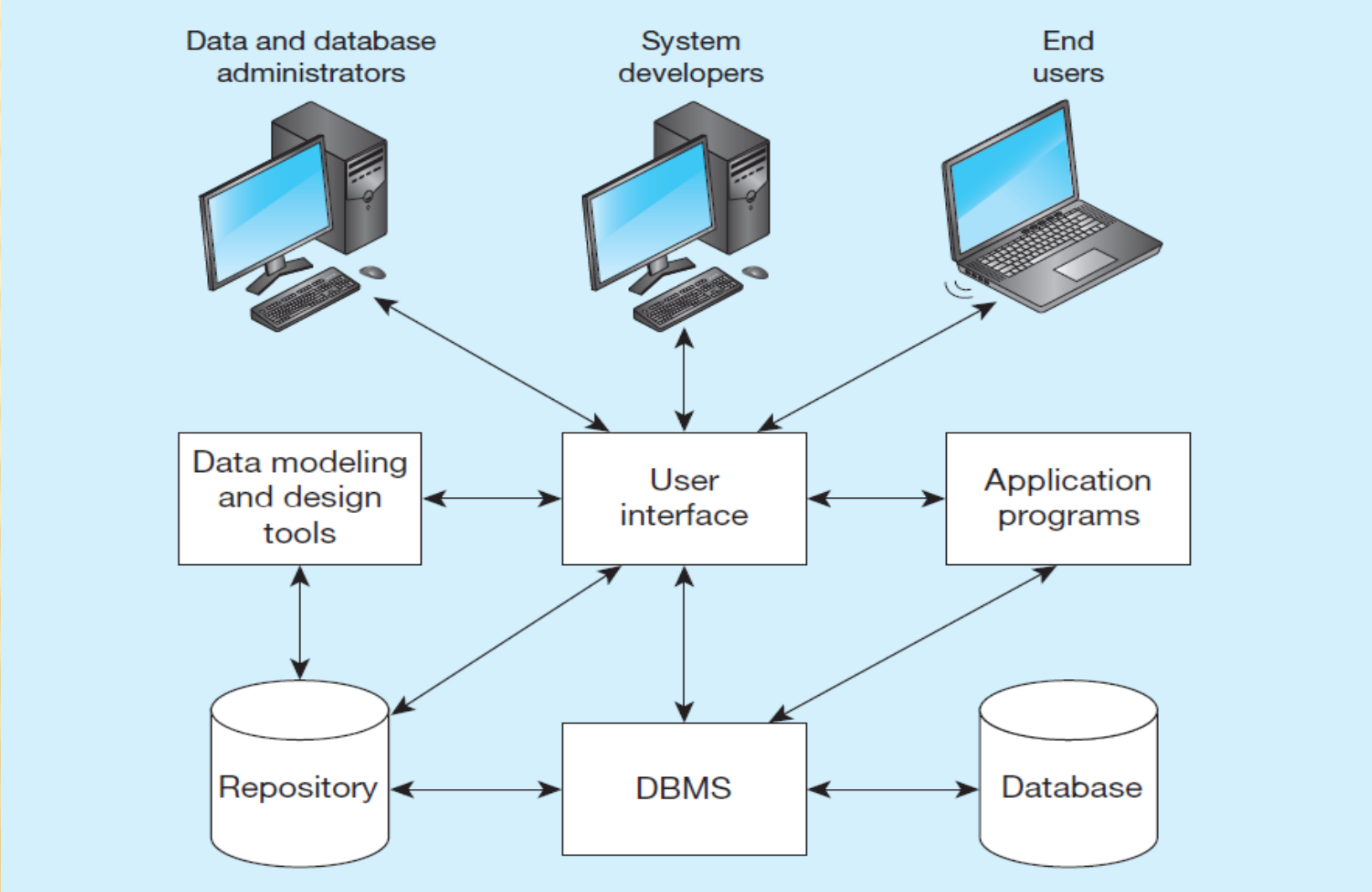
<table>
<tr><th></th><th>Operational</th><th colspan="2">Informational</th></tr>
<tr><td></td><td>Transactional</td><td>Analytical–
Data Warehousing</td><td>Analytical–
Big Data</td></tr>
<tr><td>Technology</td><td>Relational</td><td>Relational</td><td>Non-relational</td></tr>
<tr><td>Modeling</td><td>Conceptual data modeling with (E)ER (Chapters 2 and 3)</td><td rowspan="3">Data warehousing and data integration (Chapter 9)</td><td rowspan="3">Big data technologies, including Hadoop & NoSQL (Chapter 10)</td></tr>
<tr><td>Design</td><td>Logical data modeling with the relational model; Normalization (Chapter 4)</td></tr>
<tr><td>Infrastructure</td><td>Physical design of relational databases; Security; Cloud computing (Chapter 8)</td></tr>
<tr><td>Access</td><td>SQL (Chapters 5 and 6)

Applications with SQL (Chapter 7)</td><td></td><td></td></tr>
<tr><td>Data analysis</td><td colspan="3">Analytics and its implications (Chapter 11)</td></tr>
<tr><td>Governance and data management</td><td colspan="3">Lifecycle (Chapter 1)
Governance, data quality, and master data management (Chapter 12)</td></tr>
</table>

Components Of The Database Environment

- Data modeling and design tools – automated tools used to design databases and application programs
- Repository – centralized storehouse of metadata
- Database Management System (DBMS) – software for managing the database
  Access 最小 DBMS
  MS SQL 中型 DBMS
  Oracle 最大型 DBMS
- Database – storehouse of the data
- Application Programs – software using the data
- User Interface – text, graphical displays, menus, etc. for user
- Data/Database Administrators – personnel responsible for maintaining the database
  資料庫管理師
- System Developers – personnel responsible for designing databases and software
- End Users – people who use the applications and databases

Figure 1-6 Components of the Database Environment



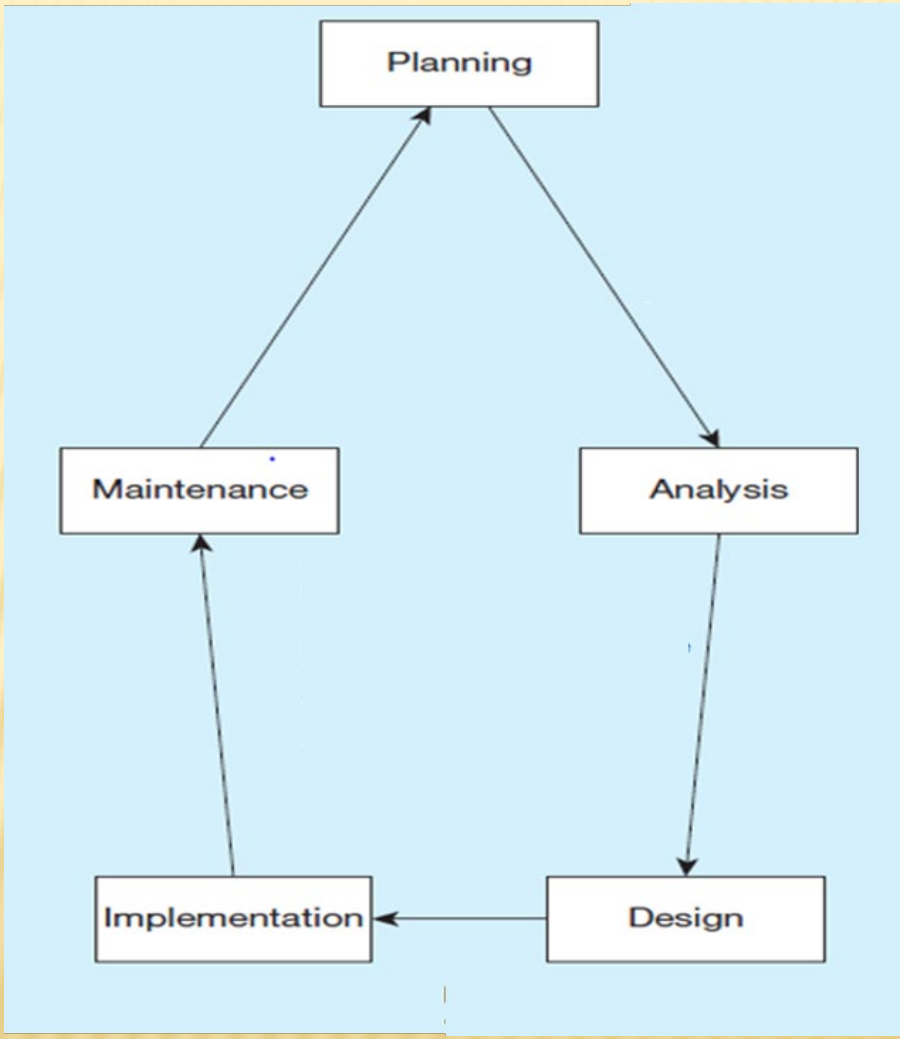
The Database Development Process

- S D L C
  - System Development Life Cycle
  - Detailed, well-planned development process
  - Time-consuming, but comprehensive
  - Long development cycle
- Prototyping
  - Rapid application development (R A D)
  - Cursory attempt at conceptual data modeling
  - Define database during development of initial prototype
  - Repeat implementation and maintenance activities with new prototype versions

Systems Development Life Cycle (SDLC)

- The traditional methodology used to develop, maintain, and replace information systems
- Five main steps:
  1. **Planning** – preliminary understanding of business situation. Enterprise model and conceptual data modeling.
  2. **Analysis** – thorough analysis of business situation, leading to functional requirements. Detailed conceptual data modeling.
  3. **Design** – logical and physical database design, to develop technology and organization.
  4. **Implementation** – writing programs, building databases, testing, installing, training, and documenting.
  5. **Maintenance** – monitoring, repairing, and enhancing.

From Figure 1-8 Database Development Activities During the SDLC



Alternative Information Systems Development Approaches

- Traditional SDLC: methodical, structured, and time consuming
- Rapid Application Development (R A D): faster and more adaptive, especially when a database is already in place
- Several flavors:
  - Prototyping
  - Agile methodologies
  - eXtreme programming
  - Scrum
  - DSDM (dynamic system development methodologies)

Figure 1-9 The Prototyping Methodology and Database Development Process

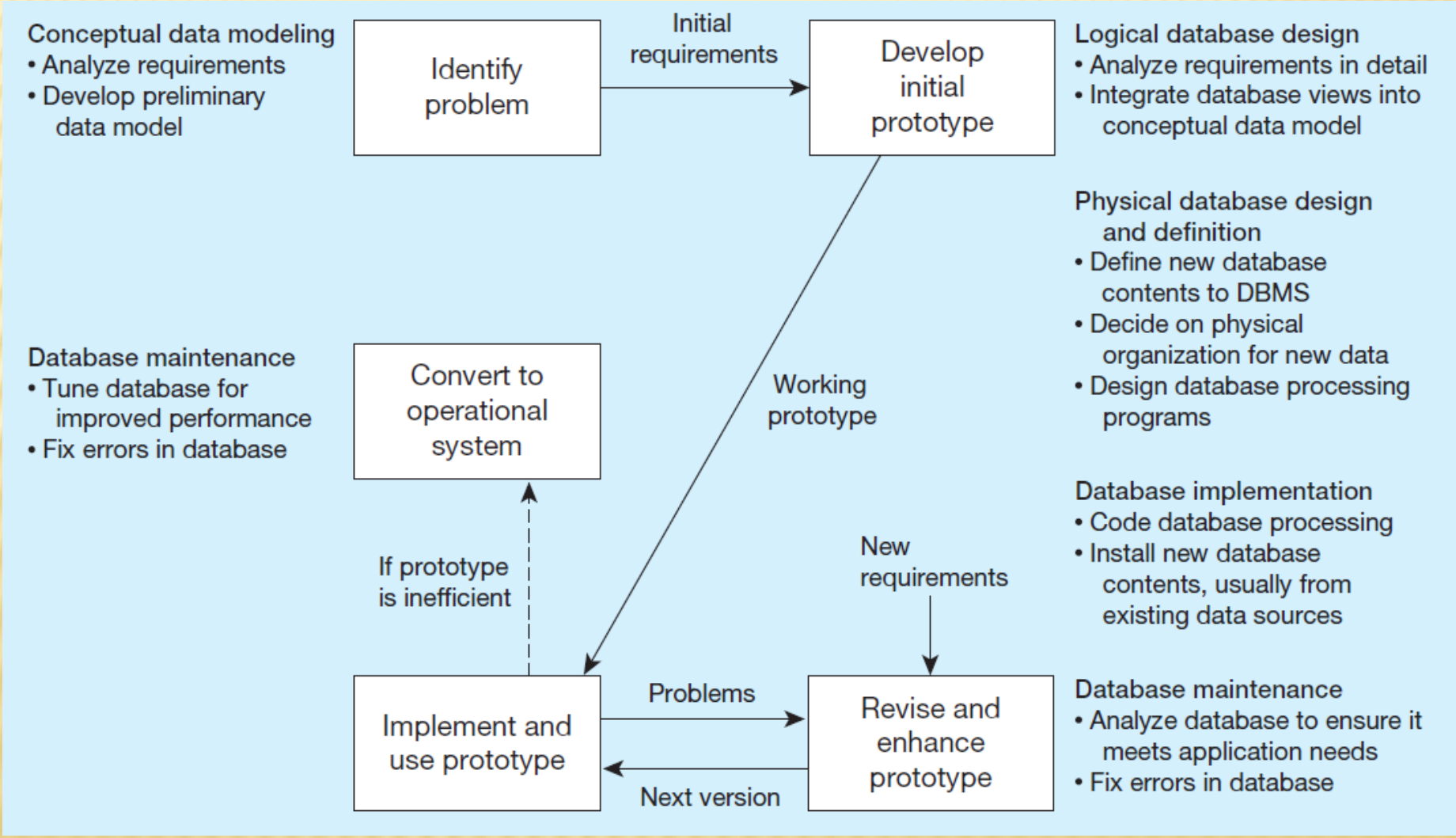
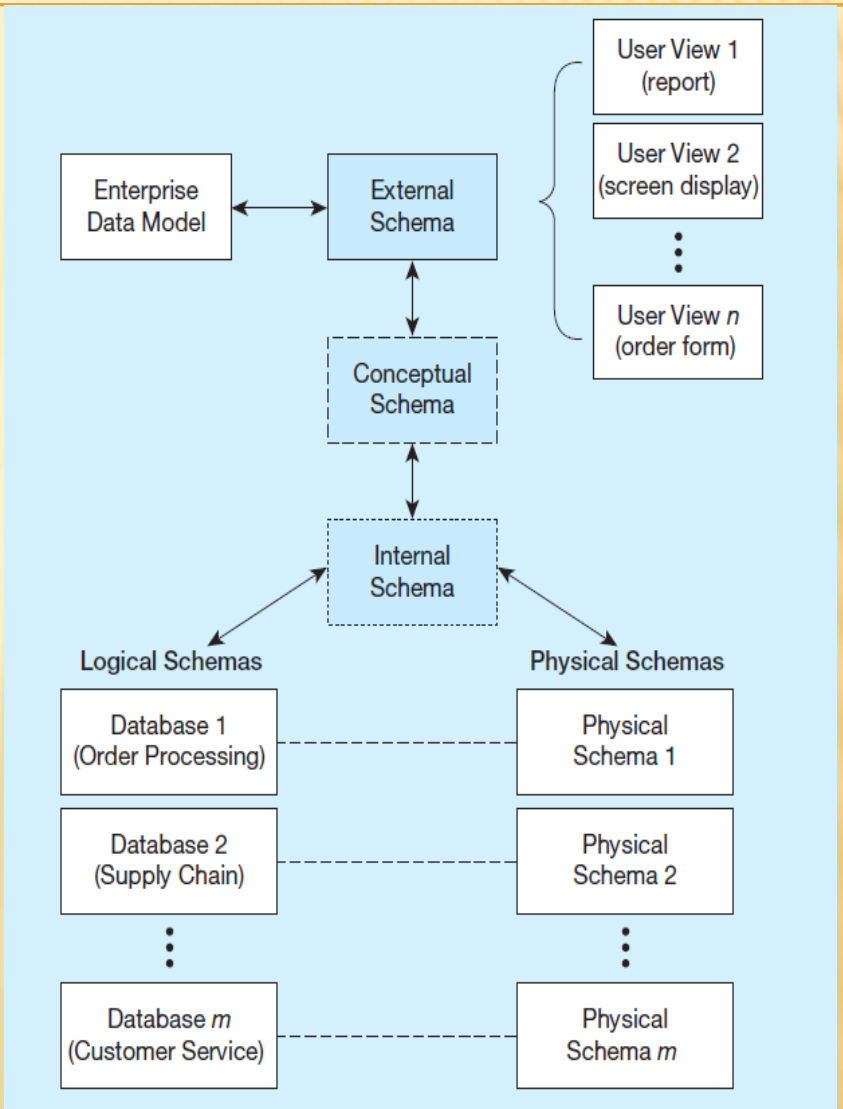


Figure 1-10 Three-schema Architecture

綱要

- Different people have different views of the database
- External Schema
  - User Views
  - Subsets of Conceptual Schema
  - Can be determined from business-function/data entity matrices
  - D B A determines schema for different users
- Conceptual Schema (E R models)
- Internal Schema (logical and physical structures)



Managing People Involved In Database Development

- Project – a planned undertaking of related activities to reach an objective that has a beginning and an end
- Initiated and planned in planning stage of SDLC
- Executed during analysis, design, and implementation
- Closed at the end of implementation

Database Project Team Members (1 OF 2)

- Business analysts – analyze business situation and establish requirements
- Systems analysts – like business analysts, but also have technical expertise for overall information systems
- Database analysts and data modelers – analysts who focus on database
- Users – the "customers" communicate their needs to analysts
- Programmers – coders of the programs that interact with the database

Database Project Team Members (2 OF 2)

- Database architects – establish standards for data in business units
- Data administrators – responsible for existing databases, ensuring data integrity and consistency
- Project managers – oversee the projects, manage the personnel
- Other technical experts – network, operating system, documentation, etc.

Evolution Of Database Systems

- Driven by four main objectives:
  - Need for program-data independence in order to reduce maintenance
  - Desire to manage more complex data types and structures
  - Ease of data access for less technical personnel
  - Need for more powerful decision support platforms

Figure 1-11 The Range Of Database Technologies: Past And Present (1 OF 4)

(a) Evolution of database technologies

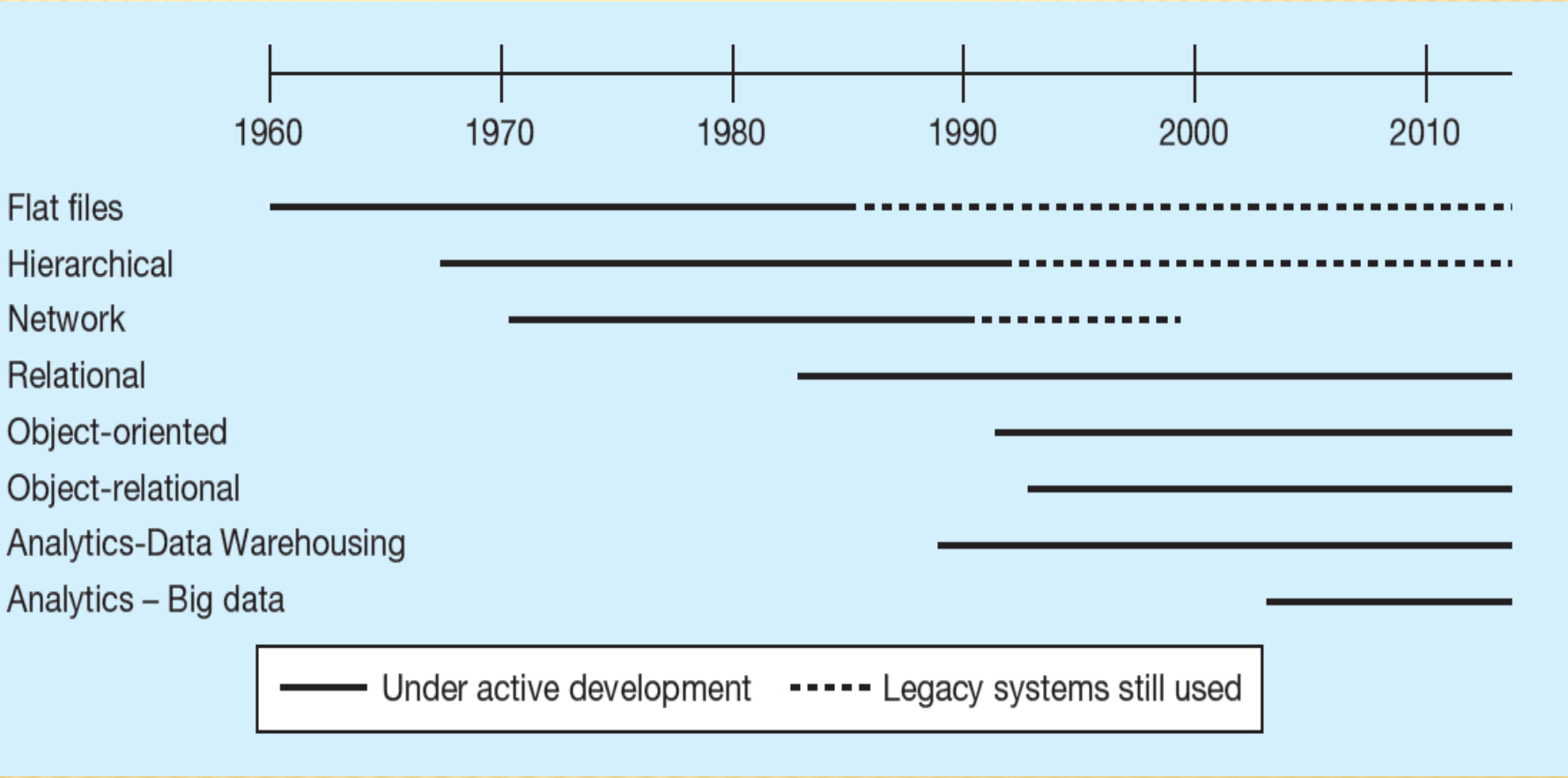


Figure 1-11 The Range Of Database Technologies: Past And Present (2 OF 4)

(b) Database architectures

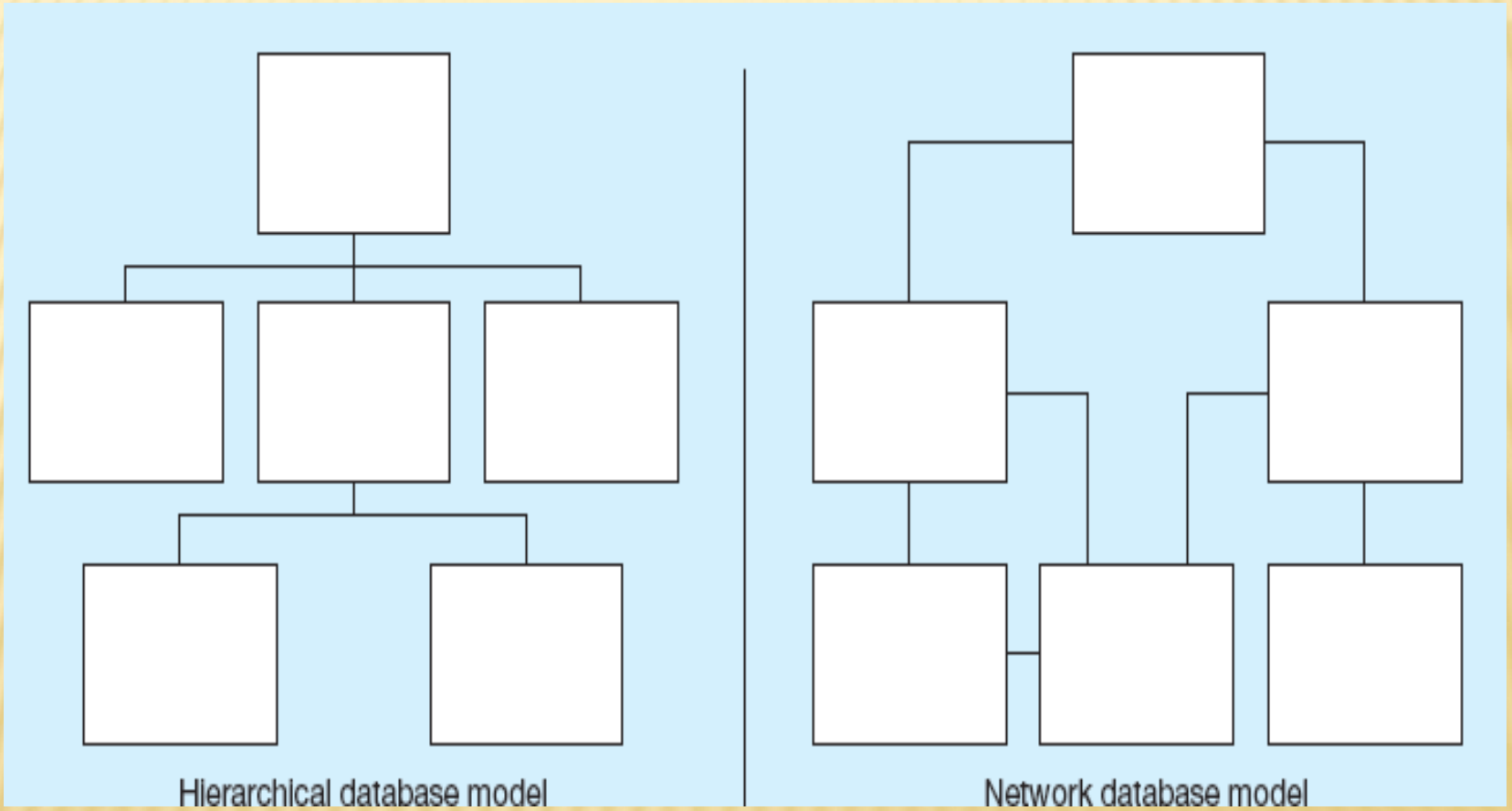


Figure 1-11 The Range Of Database Technologies: Past And Present (3 OF 4)

(b) Database architectures

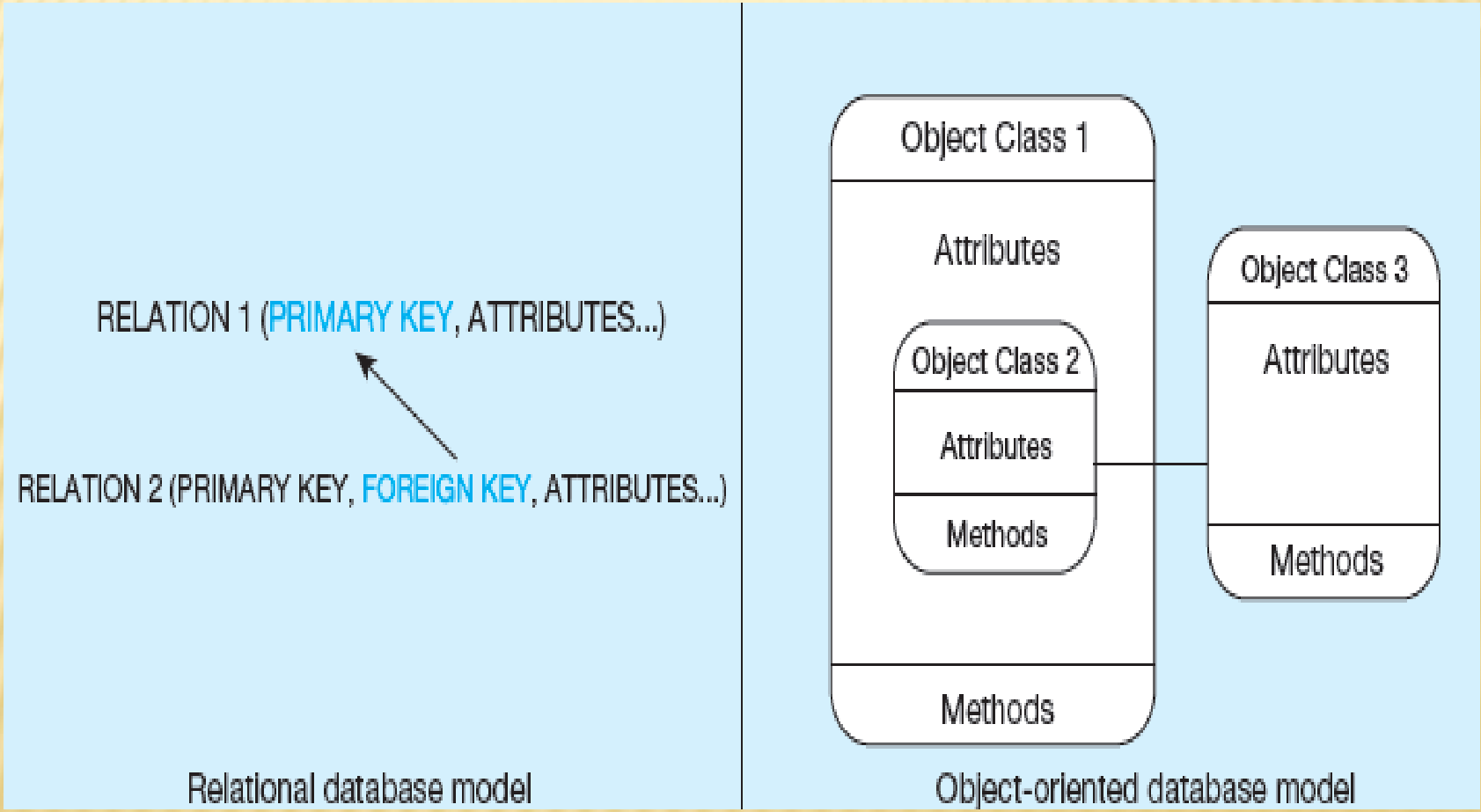
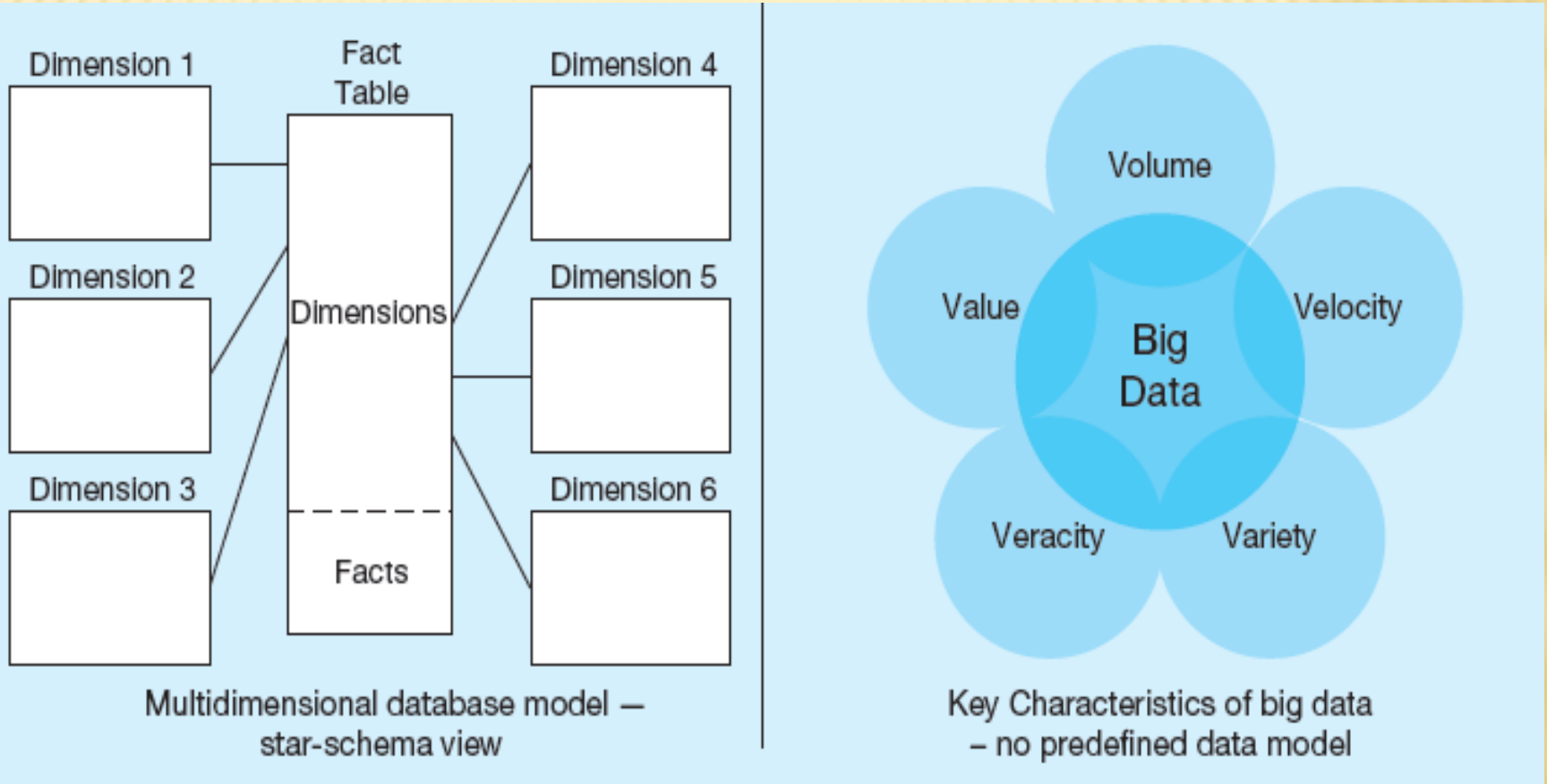


Figure 1-11 The Range Of Database Technologies: Past And Present (4 Of 4)

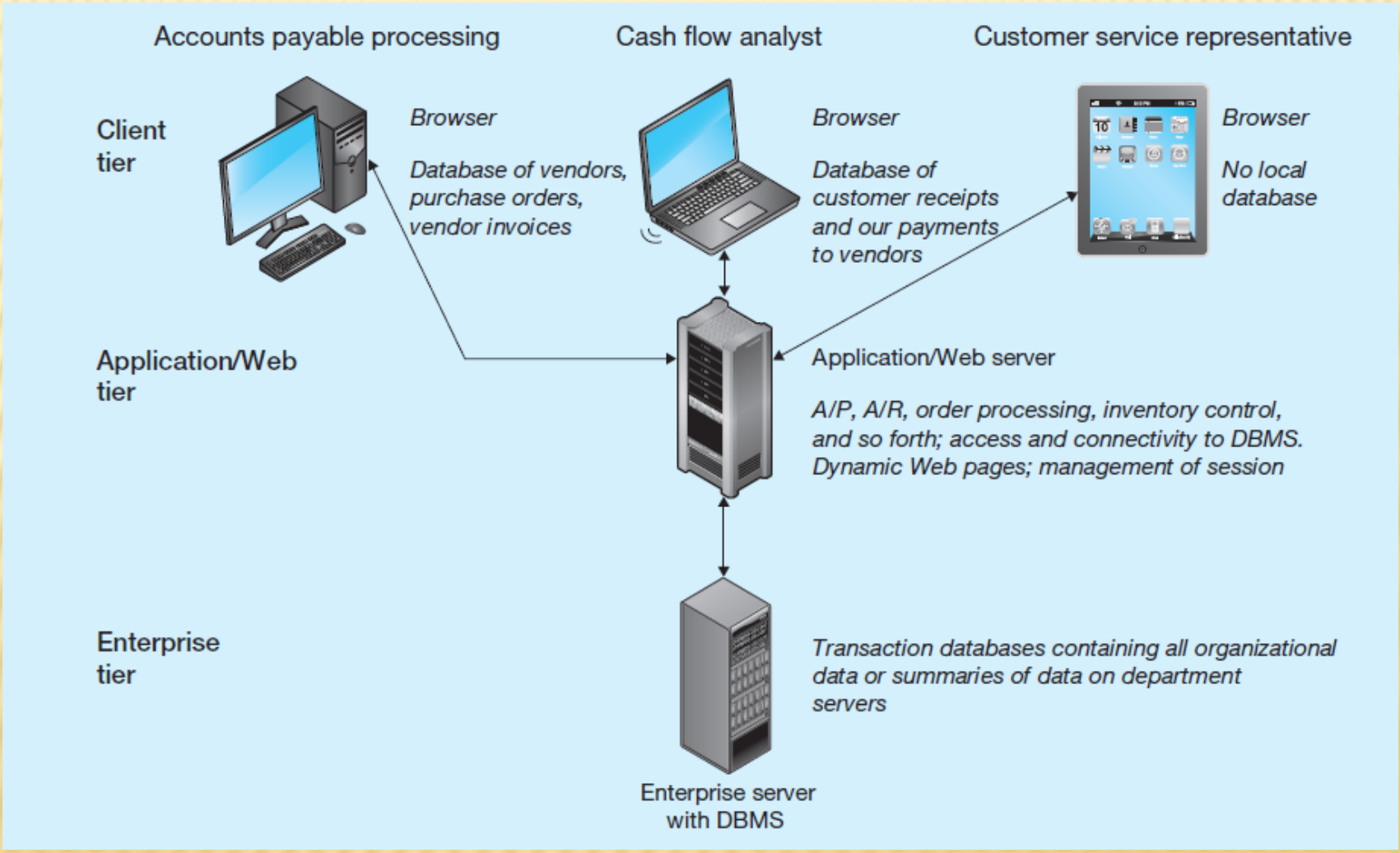
(b) Database architectures



The Range Of Database Applications

- Personal Databases
  - Typical size in the megabytes
  - Intended for one user
- Departmental Multi-Tiered Client/Server Databases
  - Typical size in the gigabytes
  - Intended for several users, usually doesn't exceed 100, department wide
- Enterprise Applications
  - Typical size in the gigabytes, terabytes, or even petabytes
  - Intended for a very large user base, company wide

Figure 1-12 Multi-Tiered Client/Server Database Architecture

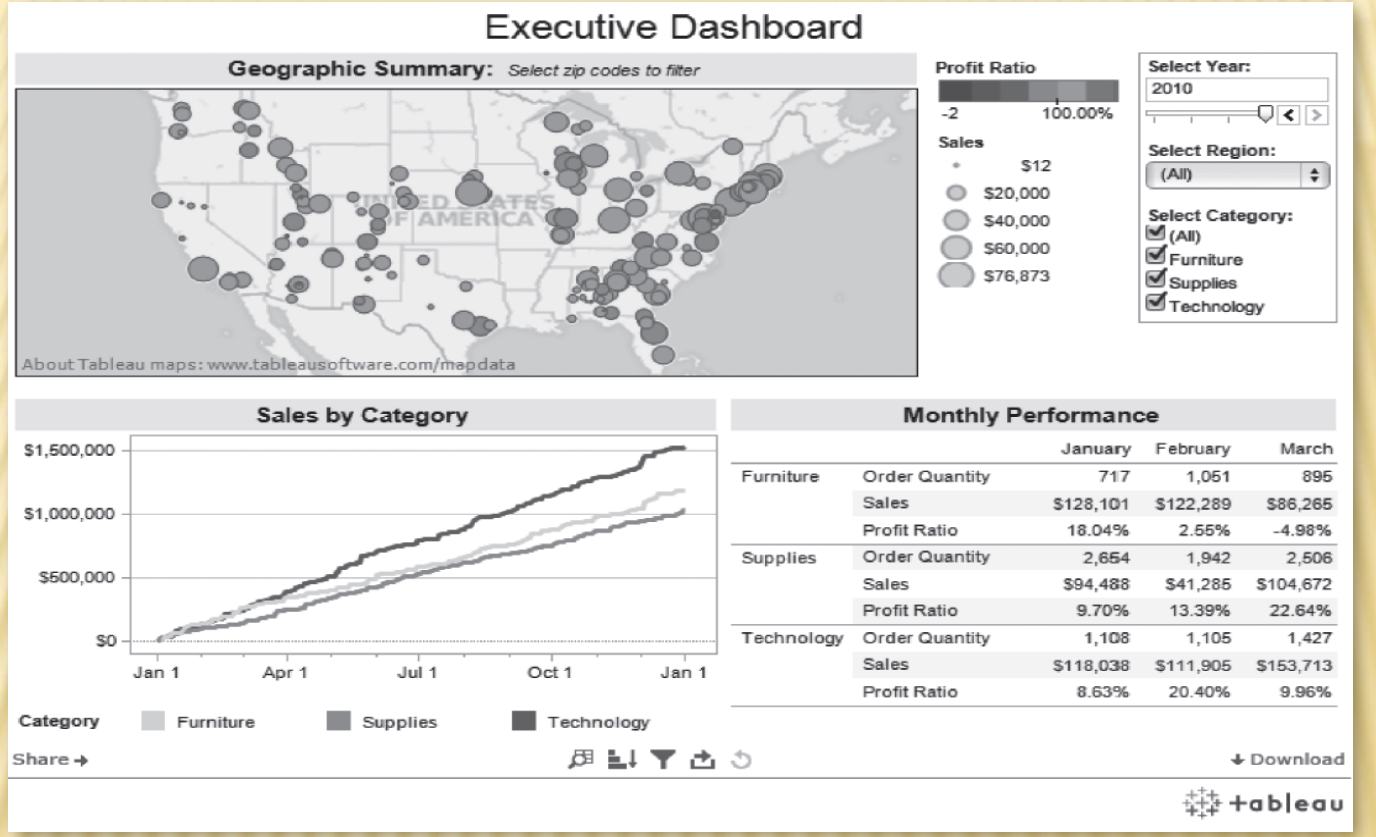


Types Of Enterprise Applications

- Enterprise Systems (typically involve relational databases)
  - Backbone of an organization
  - Enterprise resource planning (E R P)
  - Customer relationship management
  - Supply chain management
  - Human resource management and payroll
- Data Warehouses (typically involve relational databases)
  - Integrates data from multiple data sources
  - Maintain historical data
  - Help identify patterns and trends
- Data Lakes (often don't involve relational databases)
  - Large integrated repository for internal and external data that does not follow a predefined schema

企業資源規劃

Figure 1-13 An Example of an Executive Dashboard



(http://public.tableausoftware.com/profile/mirandali#!/vizhome/Executive-Dashboard_7/ExecutiveDashboard)

Courtesy Tableau Software

PINE VALLEY FURNITURE (PVF)

- PVF is a fictional furniture company
- Will be used as a case throughout the book
- Chapter 1 topics include
  - Database Evolution at Pine Valley Furniture Company
  - Project planning
  - Analyzing database requirements
  - Designing the database
  - Using the database
  - Administering the database
  - Future of databases at PVF